

INDUSTRIAL EQUIPMENT MONITORING PLATFORM

Scope, End-to-End Demo Path, Success Criteria and Known Limitations

Product Owner / Project Manager Report

Project	Industrial Equipment Monitoring Platform
Release	Portfolio MVP
Report role	Product Owner / Project Manager
Project owner	Jerry
Report date	23 September 2026
Status	Scope baseline for team review and execution
Source baseline	Requirements Specification v1.0 + Sprint 01 QA and Technical reports

Purpose: establish an execution-ready scope boundary for the portfolio MVP, document the intended end-to-end path, define what the demonstration must prove, and make known limitations explicit so delivery decisions remain consistent across firmware, gateway, backend, frontend, QA and DevOps.

Document rule: the Requirements Specification remains the governing baseline. This report clarifies scope and demo acceptance; it does not silently remove baseline requirements. Any conflict or change requires documented change control.

Contents

1. Executive Summary
 2. Source Basis and Scope Authority
 3. Scope Statement and Product Boundary
 4. In-Scope End-to-End Path
 5. In-Scope MVP Capabilities
 6. Explicit Out-of-Scope Items
 7. Demo Success Criteria
 8. Known Limitations and Assumptions
 9. Open Decisions and Dependencies
 10. Delivery Risks and Mitigations
 11. Demonstration Runbook and Evidence
 12. Team Ownership and Responsibilities
 13. Scope Change Control and Release Gate
- Appendix A. Requirement Traceability Matrix

1. Executive Summary

The Portfolio MVP is an advisory industrial equipment monitoring platform. Its job is to demonstrate a credible end-to-end telemetry workflow: acquire or simulate temperature readings, move them through local MQTT and a Linux gateway, persist them reliably, ingest them through FastAPI, store them in PostgreSQL, and present operational status, history and alerts in a React dashboard.

The primary product story is failure-aware rather than “happy path only”: a high-temperature condition becomes visible to an operator; the backend is then unavailable for ten minutes while the local path remains available; the gateway buffers committed readings; after connectivity returns, the backlog is replayed without duplicate database records. This is the core portfolio proof of engineering depth across embedded, edge, backend, database and frontend boundaries.

This report also protects the MVP from uncontrolled expansion. Production-grade identity management, high availability, complete enterprise security hardening and production-scale performance commitments are not demo-scope deliverables. However, basic MVP authentication/authorization and the minimum security/reliability behaviors already present in the requirements baseline are still expected; “out of scope” must not be interpreted as permission to expose an unauthenticated or obviously unsafe system.

Traceability: Requirements Specification pp. 1-2, Product purpose, Scope and assumptions, Proposed implementation baseline; pp. 6-7, acceptance scenarios and milestones.

2. Source Basis and Scope Authority

This report is derived from three project documents and uses the Requirements Specification v1.0 as the governing source. The Sprint 01 QA and Technical reports are supporting interpretations and decision logs. Where they contain proposals or open questions, those items are not treated as approved requirements unless explicitly called out as a PO/PM decision in this report.

Source	Authority in this report	Key contribution
Requirements Specification v1.0	Governing baseline	Product purpose, MVP scope, architecture baseline, functional/non-functional/security requirements, acceptance scenarios, milestones and Definition of Done.
QA Sprint 1 - Telemetry Contract, Decision Log and Open Questions	Supporting QA interpretation	Deterministic event identity, timestamp separation, mixed-batch behavior, test conditions and unresolved contract questions.
Sprint 01 Technical Report	Supporting technical interpretation	Contract-first M1 vertical slice, accepted technical decisions, gateway queue lifecycle, timestamp ownership and API contract examples.

Scope precedence: Requirements baseline > approved change record > this PO/PM clarification > sprint working-document proposals. A working-document proposal does not automatically become a product requirement.

3. Scope Statement and Product Boundary

3.1 Product objective

Build a portfolio-quality monitoring platform that lets maintenance-oriented users observe temperature, detect stale/disconnected devices, review historical readings, and acknowledge alert episodes. The MVP demonstrates software integration and reliability behaviors; it is not an automatic control or safety system.

3.2 Physical and simulated boundary

- Physical demonstration: one ESP32, one temperature sensor and one Raspberry Pi gateway.
- Simulator: additional devices and deterministic test events used to exercise volume, duplicates, invalid payloads, replay and ordering cases.
- Organisation model: one organisation with multiple sites; users are granted access to specific sites.
- Primary metric for MVP: temperature in celsius.
- Monitoring is advisory only; no equipment actuation, emergency shutdown or safety certification.

Traceability: Requirements Specification §2, p.1; telemetry contract §6, pp.3-4.

3.3 Scope principle

PO/PM scope principle: prove a complete, explainable vertical product story before adding enterprise breadth. Reliability and authorization semantics take priority over optional features if schedule pressure occurs.

4. In-Scope End-to-End Path

4.1 Target demo architecture

The requested end-to-end path is in scope, with one important architectural clarification: the React application must not connect directly to PostgreSQL. The browser consumes backend APIs; FastAPI owns database access and authorization. The logical path is therefore:

Sensor / Simulator -> MQTT -> Linux Gateway + SQLite -> HTTPS Batch -> FastAPI -> PostgreSQL
React Dashboard -> FastAPI query APIs -> PostgreSQL

Stage	Technology	Responsibility	Key boundary / output
1. Sensor	ESP32, ESP-IDF, C/C++	Acquire real temperature and form versioned telemetry.	Publishes device telemetry to MQTT; default real reading interval is five seconds.
1b. Simulator	Deterministic test tool	Generate additional valid/invalid/duplicate/out-of-order device events.	For full demo-path tests it should publish MQTT-formatted telemetry; for Sprint 01 contract tests it may post directly to the ingestion API.
2. Local messaging	Mosquitto, MQTT QoS 1	Transport local telemetry from devices to gateway.	At-least-once delivery means duplicates are possible; broker acknowledgement is not a durability guarantee.
3. Linux gateway	Raspberry Pi, C++17, SQLite	Validate, timestamp receipt, persist before forwarding, batch and retry.	Durability guarantee begins after SQLite commit. Accepted/duplicate events leave queue; permanent rejects go to bounded quarantine; transient failures stay queued.
4. Backend ingestion	FastAPI	Authenticate gateway, validate device ownership/schema, classify item results, enforce idempotency.	POST /api/v1/telemetry/batches is the Sprint 01 proposed contract. Accepted outcome is returned only after DB commit.
5. Persistence	PostgreSQL	Store registry, telemetry, current state, alerts/audit data.	Unique telemetry identity: (device_id, boot_id, sequence_number).
6. Dashboard	React + TypeScript	Show current device state, history and alert workflow.	Consumes authorized FastAPI endpoints; displays latest temperature, age, connection state, active alerts and historical charts.

Traceability: Requirements Specification §4, p.2; FR-04 to FR-12, pp.2-3; telemetry processing rules p.4. Sprint 01 Technical Report §2, pp.3-4 and §5-6, pp.5-6.

4.2 Normal telemetry flow

- ESP32 or simulator creates a telemetry event with schema_version, device_id, boot_id, sequence_number, measurement/uptime fields, metric, value, unit and quality metadata.
- MQTT transports the event locally. QoS 1 can redeliver, so every downstream component must tolerate duplicates.
- Gateway receives the event, stamps gateway_received_at, validates enough to protect its queue, and commits the event to SQLite before treating it as recoverable.
- Gateway sends one or more queued events to FastAPI as a bounded batch over HTTPS.
- FastAPI derives gateway identity from its credential, verifies device ownership, validates each item, detects duplicate/identity-conflict cases, and commits valid events to PostgreSQL.
- Backend returns deterministic per-item outcomes only after commit. Gateway removes accepted/matching-duplicate events, quarantines permanent invalid events, and retains transient failures for retry.
- React retrieves current state/history/alerts through FastAPI and presents the operator-facing view.

4.3 Outage and recovery flow

The primary failure path is a ten-minute backend or internet interruption while local MQTT remains available. The gateway continues committing received telemetry to SQLite. The queue grows, survives gateway process restart if necessary, and is replayed after the backend returns. Duplicate protection is critical because the gateway may retry after an uncertain or lost HTTP response even if the backend already committed the data.

Critical recovery rule: an HTTP/API “accepted” result is valid only after the PostgreSQL transaction commits. If the response is lost after commit, retrying the same event identity must produce a duplicate outcome rather than a second stored row.

4.4 Sprint 01 versus full MVP path

Sprint 01 intentionally proves a narrower contract-first path: deterministic simulator -> FastAPI ingestion -> PostgreSQL persistence and verification. The full physical ESP32 -> MQTT -> Raspberry Pi gateway -> backend -> dashboard path is a Week 2 physical-demonstration milestone. This distinction prevents M1 from being delayed by hardware/UI work while still preserving the final architecture contract.

Traceability: QA Sprint 1 §1, pp.1-2; Sprint 01 Technical Report §1-2, p.3; Requirements Specification §11, p.7.

5. In-Scope MVP Capabilities

Capability	MVP scope statement
Registry	Sites, gateways and devices can be registered; device IDs are unique and device-to-gateway/site relationships are authoritative.
MVP authentication and authorization	Users authenticate; backend enforces role/site access. Gateway credentials are separate and device ownership is checked.
Telemetry	Real ESP32 temperature readings plus simulated devices; explicit sensor error instead of fabricated temperature.
Durable edge queue	Gateway persists before forwarding; retry and bounded backoff support temporary backend failure.
Idempotent ingestion	Stable event identity; same event may be resent safely; identity reuse with changed immutable content is rejected/logged.
Mixed-batch validation	Valid, duplicate and invalid items receive independent outcomes; invalid items do not indefinitely block valid delivery.
Current status	Latest temperature, reading age, connection state and active alert count; unavailable data is explicit.
History	Device/time-range history using trustworthy measurement time and visible data gaps.
Connection state	Stale connection detection based on heartbeat/timeout behavior; historical replay must not mark device online.
Temperature alert	Configurable high-temperature rule, one alert episode per condition, acknowledgement separate from physical recovery.
Operational evidence	Logs, health/queue metrics, reproducible setup, CI evidence, backup/restore procedure and known-limitations list.

6. Explicit Out-of-Scope Items

The following items are excluded from the current demo/MVP scope or from its production-readiness claim. The exclusions are intentional scope boundaries, not defects. Minimum controls already required by the baseline remain in force.

Out-of-scope area	Excluded	What still remains in scope
Production identity and access management	Enterprise SSO/federation, MFA, corporate IdP integration, automated joiner/mover/leaver flows, sophisticated session/device trust, external customer IAM.	Basic application authentication plus role/site authorization remains in scope.
High availability and scale-out architecture	Kubernetes, multi-node active-active services, broker/database clusters, multi-region design, automatic failover, zero-downtime deployment and autoscaling.	Docker/Compose deployment is sufficient for MVP; restart recovery at gateway is still in scope.

Out-of-scope area	Excluded	What still remains in scope
Full security hardening / certification	Formal penetration test, ISO/SOC certification, full SIEM/SOC integration, enterprise secret rotation infrastructure, WAF program, exhaustive threat-model closure, compliance certification.	Baseline HTTPS/TLS, password hashing, secret hygiene, request/login limits, authorization tests, safe errors/logs and dependency review remain expected.
Full production performance commitments	24x7 SLO/SLA, production capacity guarantee, multi-day soak, internet-scale load, autoscaling proof and formal performance certification.	Baseline NFR release-test targets remain tracked for later evidence; the short product demo itself is not a production benchmark.
Automatic control and safety functions	Equipment actuation, emergency shutdown, safety interlocks and safety certification.	Monitoring and alerts are advisory only.
Advanced analytics	Predictive maintenance, anomaly detection and AI/ML recommendations.	Threshold-based temperature alerting only.
Additional industrial protocols	Modbus, OPC UA, CAN gatewaying and other fieldbus/industrial protocol integrations.	MQTT is the MVP device-to-gateway transport.
External notifications	Email, SMS, paging and escalation schedules.	Dashboard alert workflow and acknowledgement only.
Enterprise platform features	Customer self-service, billing and enterprise SSO.	Single-organisation portfolio MVP.
Fleet firmware management	OTA firmware rollout, staged deployment and rollback for device fleet.	Firmware is manually managed for demo hardware.
ESP32 offline flash buffering	Persistent sensor-side backlog when the gateway/MQTT path is unavailable.	Recovery guarantee begins after the gateway has committed the reading.
AWS production deployment	Cloud production deployment and cloud operationalization.	AWS is an extension after MVP acceptance tests pass.

Traceability: Requirements Specification §2, p.1; §8-9, pp.4-6; §11, p.7. User-requested out-of-scope clarification applied to production-grade auth, HA, security hardening and performance commitments.

7. Demo Success Criteria

The demo is successful when it proves the product story end to end and produces evidence that can be reviewed later. The criteria below separate core demonstration gates from broader release evidence.

7.1 Core demo pass criteria

ID	Scenario	Action	Pass condition
DS-01	System startup and health	From documented setup, MQTT, gateway, FastAPI, PostgreSQL and React start successfully; required test device/site/gateway records exist.	Services healthy; dashboard loads; no blocking errors.
DS-02	Real sensor visibility	Publish a physical ESP32 temperature reading through MQTT and gateway.	Correct device temperature, timestamp/unit and device status become visible in the dashboard.
DS-03	Simulator equivalence	Publish simulator telemetry using the same device contract (preferably through MQTT for the full demo path).	Simulated device appears with correctly attributed telemetry; no path-specific schema fork.
DS-04	Idempotent storage	Retry an identical event multiple times.	Exactly one PostgreSQL telemetry row exists for the event identity; retries are classified as duplicates.
DS-05	High-temperature alert	Send three consecutive valid readings above 30 C.	Exactly one active alert opens; later high readings do not create duplicate active episodes.
DS-06	Acknowledge and recover	Acknowledge the alert, then send three consecutive valid readings below 28 C.	Acknowledgement user/time is retained; condition resolves only after recovery readings.
DS-07	Ten-minute outage buffering	Disconnect backend/internet for ten minutes while local MQTT remains available.	Gateway queue grows; committed events survive locally; no manual gateway restart is required for normal recovery.

ID	Scenario	Action	Pass condition
DS-08	Backlog replay without duplication	Restore backend connectivity.	Queued events are uploaded; database counts are correct; duplicate retries do not create additional rows; older replay does not replace newer current state.
DS-09	Gateway restart durability	Restart gateway during backend outage.	Previously committed queue survives and forwarding resumes after recovery.
DS-10	Stopped device versus replay	Stop live device while replaying historical backlog.	Device becomes offline according to stale-connection rules; historical replay does not make it appear online.

7.2 Contract and error-behavior criteria

Condition	Required observable result
Same identity + same immutable content	duplicate; no second stored record; safe to remove from gateway queue.
Same identity + different immutable content	explicit identity conflict rejection; original row unchanged; conflict logged.
Mixed valid / duplicate / invalid batch	every item receives deterministic outcome; valid items can succeed independently.
Lost response after backend commit	gateway retry is safe; already committed events return duplicate and row counts remain unchanged.
Unknown device / wrong gateway / invalid schema/unit/value	explicit permanent rejection reason; event does not retry forever.
Transient timeout / 429 / 5xx	event remains queued and retry behavior is bounded.

7.3 Demo completion evidence

- Short video or live demo showing normal operation, failure and recovery.
- Screenshots/log extracts showing sensor reading, alert state, acknowledgement and recovery.
- Gateway queue evidence before, during and after outage/restart.
- Database query/test evidence proving idempotency and no duplicate stored event for retries.
- Acceptance-test results linked to requirement/scenario IDs.
- Known limitations and unresolved decisions shown explicitly in the release notes.
- Actual performance numbers, if presented in a CV/portfolio, must come from measured tests rather than the baseline targets.

Traceability: Requirements Specification §10, p.6; §12, p.7; Technical Report §11, pp.9-10.

8. Known Limitations and Assumptions

ID	Limitation / assumption	Impact
KL-01	Single-organisation MVP	Architecture and authorization model are not yet validated for true multi-tenant SaaS isolation.
KL-02	Small physical hardware footprint	Only one ESP32, one temperature sensor and one Raspberry Pi gateway are required for the physical demo; simulator represents additional devices.
KL-03	Temperature-only telemetry	MVP metric is temperature in celsius; generic multi-metric/device-class support is not proven.
KL-04	At-least-once MQTT delivery	QoS 1 can produce duplicate deliveries. Application idempotency is mandatory; a broker acknowledgement does not prove SQLite persistence.
KL-05	Recovery guarantee starts at gateway commit	A reading can be lost before it is committed to gateway storage; ESP32 flash buffering is deferred.
KL-06	No production HA	A single broker/gateway/backend/database instance may be a single point of service interruption. Restart durability is tested, not clustered failover.

ID	Limitation / assumption	Impact
KL-07	Demo-grade identity/security posture	Minimum baseline controls apply, but the system is not claiming enterprise identity integration, certification or complete production hardening.
KL-08	Performance claims limited to measured evidence	Baseline NFR values are test targets. Until executed and recorded with hardware/software/network conditions, they are not achieved results.
KL-09	Historical replay is not live state	Old/stale/out-of-order data can populate history but must not advance live alert state or make an offline device appear online.
KL-10	Clock trust requires explicit handling	measured_at can be null or untrusted. Device, gateway and backend time concepts must remain separate; any unresolved clock-quality semantics must be visible in the decision log.
KL-11	Bounded queue/quarantine details may require final configuration	The requirements demand bounded behavior, but exact production sizing/retention values are implementation/configuration choices and must be documented.
KL-12	Advisory monitoring only	No automatic control, emergency shutdown or safety assurance is provided.

9. Open Decisions and Dependencies

The following items should be closed, explicitly deferred, or linked to an owner before implementing the full alert/reliability workflow. They come from QA/Technical sprint working documents and are included here to prevent hidden implementation assumptions.

Decision	Question	Owner(s)	Needed by
Clock quality model	Define supported reading/clock-quality values and what makes measured_at trustworthy.	Firmware + Gateway + Backend + QA	Before final history/alert age logic.
Cross-boot live ordering	Define how a new boot supersedes an old boot so delayed old-boot telemetry cannot replace new live state.	Backend Lead / Architect	Before full alert engine.
Alert age/gap timeline	Confirm timestamp used for >30 s stale-event test and >15 s pending-count gap rule.	Backend Lead + QA	Before M4 alert implementation.
Rule scope	Decide whether temperature rule is per device, site, class or another scope.	PO + Backend Lead	Before admin alert-rule UI/API is finalized.
Reboot during pending alert	Confirm whether reboot resets pending consecutive count.	PO + Backend Lead + QA	Before alert-state tests.
Batch HTTP semantics	Define HTTP status for syntactically valid mixed-outcome batches and stable machine-readable error codes.	Backend Lead + Gateway Developer	Before gateway production retry logic.
Quarantine bound	Choose bounded size/retention policy for permanently invalid gateway events.	Gateway Lead + PO/PM	Before M3 reliability sign-off.
Concurrent same-device batches	Define serialization/locking/idempotent live-state update strategy.	Backend Lead / Architect	Before concurrent alert processing.

PO/PM action: open decisions are not reasons to guess in code. Each decision must be recorded with the affected requirement IDs and test impact before the dependent implementation is considered Done.

10. Delivery Risks and Mitigations

Risk	Level	Why it matters	Mitigation
R-01 Scope expansion	High	UI polish, cloud, extra protocols or enterprise auth can consume time before core reliability works.	Freeze the demo path; defer extensions before weakening reliability/authorization criteria.
R-02 Contract divergence	High	Firmware, gateway and backend may interpret identity/timestamps differently.	Version telemetry contract; shared examples; QA contract tests; approved decision log.

Risk	Level	Why it matters	Mitigation
R-03 False reliability claim	High	MQTT receipt could be mistaken for durable capture.	Define recovery boundary at gateway SQLite commit and document pre-commit loss limitation.
R-04 Duplicate records on retry	High	Lost responses or QoS 1 redelivery can create duplicate data if identity is weak.	DB uniqueness on composite identity plus deterministic duplicate/conflict behavior.
R-05 Replay corrupts live state	High	Backlog could overwrite latest value, alert state or online status.	Separate history persistence from live-state progression; stale/out-of-order/duplicate exclusions.
R-06 Demo environment instability	Medium	Multiple services/hardware may fail for unrelated reasons.	Scripted startup, health checks, seeded registry, pre-demo smoke test and rollback plan.
R-07 Security scope misunderstanding	Medium	“Full hardening out of scope” could be interpreted as “security optional”.	Keep baseline authz/TLS/secret hygiene/security tests as minimum controls; label only production-depth items as deferred.
R-08 Unverified performance claims	Medium	Targets could be repeated as achieved results.	Record hardware, software, dataset, network and actual results; use measured numbers only in portfolio/CV claims.

11. Demonstration Runbook and Evidence

11.1 Pre-demo checklist

- Clean checkout or known release commit identified; environment variables/secrets loaded outside source control.
- PostgreSQL migrations applied; site, gateway and demo device registry verified.
- Mosquitto, gateway, FastAPI/background worker and React services healthy.
- ESP32 publishes normal temperature and simulator can generate deterministic scenarios.
- Dashboard time zone is visible; device status and history view load.
- Test alert rule configured using demo values: trigger >30 C for three valid readings; recovery <28 C for three valid readings.
- Gateway queue/oldest-age/retry observability available for outage section.
- Database verification query/test prepared for event counts and identity checks.

11.2 Recommended three-minute portfolio demo sequence

Time	Segment	Evidence to show
0:00-0:30	Architecture + normal reading	Show physical sensor -> MQTT -> gateway -> API -> DB -> dashboard. Point out that React uses API, not direct DB access.
0:30-1:10	Alert workflow	Raise temperature/simulate three high readings, show single alert, acknowledge it, then demonstrate recovery readings.
1:10-2:15	Failure and durability	Block backend connectivity; show gateway queue growth; optionally restart gateway and show queue retained.
2:15-2:45	Recovery and idempotency	Restore backend; show backlog drain and database count consistency / no duplicates.
2:45-3:00	Close with engineering boundaries	Show device offline/replay behavior if time permits, then state known limitations and deferred production features.

11.3 Evidence package

For each demo criterion, store enough evidence that a reviewer can reproduce or inspect the result without relying on narration. Evidence may include automated test output, selected logs, database counts, screenshots and a short demo recording. The final release should link evidence to requirement or acceptance IDs.

12. Team Ownership and Responsibilities

Role	Accountability
Product Owner / PM	Own scope, priority, product decisions, success criteria, change control, demo narrative, release go/no-go.
Technical Lead / Architect	Own architecture consistency, contract decisions, cross-component interfaces, ADRs and technical risk closure.
Firmware Developer	ESP32 sensor acquisition, telemetry identity/clock metadata, MQTT publication and sensor error behavior.
Gateway Developer	MQTT subscription, SQLite durability, batching, retries/backoff, quarantine, queue metrics and recovery.
Backend Developer	FastAPI registry/ingestion/query/auth, PostgreSQL migrations, idempotency, alert/live-state logic and audit behavior.
Frontend Developer	React device status, history, alert acknowledgement UI and explicit loading/empty/error states.
QA Engineer	Acceptance matrix, contract tests, failure injection, cross-site/security checks and evidence package.
DevOps / Platform	Docker/Compose, CI, environment configuration, deployed TLS, backup/restore procedure and health/observability support.

13. Scope Change Control and Release Gate

13.1 Change control

Any change to the baseline path, telemetry contract, acceptance criteria or deferral list must record: affected requirement IDs, reason for change, acceptance-test impact, schedule impact, owner and approval. A change is not complete when only the code changes; contracts, tests and documentation must move together.

13.2 Demo go/no-go gate

Gate	Area	Decision condition
GO-1	Core path	DS-01 through DS-04 pass; real telemetry reaches dashboard and storage behavior is deterministic.
GO-2	Alert story	DS-05 and DS-06 pass with one alert episode, preserved acknowledgement and correct recovery.
GO-3	Failure/recovery	DS-07 through DS-09 pass or any omitted criterion is explicitly declared as not yet complete rather than hidden.
GO-4	Live-vs-history integrity	DS-10 or equivalent test proves replay does not falsely restore online/live state.
GO-5	Blocking defects	No blocking defect remains in the demo path.
GO-6	Evidence and limitations	Demo evidence and known limitations are published with the release candidate.
GO-7	Claim discipline	Portfolio/CV wording uses measured results only; baseline targets are not represented as achieved benchmarks.

Release decision rule: a polished UI does not compensate for broken durability, duplicate handling or authorization boundaries. When time is constrained, remove extensions before relaxing the reliability and authorization criteria.

Appendix A. Requirement Traceability Matrix

Source area	Reference	Used for	Report section
Product purpose / primary demo	Requirements §1	End-to-end workflow and ten-minute outage/recovery without duplicate records.	Sections 1, 4, 7, 11

Source area	Reference	Used for	Report section
MVP scope/deferred features	Requirements §2	Physical hardware boundary, simulator, advisory-only scope, deferred capabilities.	Sections 3, 6, 8
Implementation baseline	Requirements §4	ESP32, MQTT, Raspberry Pi/C++/SQLite, FastAPI/PostgreSQL, React/TypeScript, Docker/CI.	Section 4
FR-04/05	Requirements §5	Real readings; stable event identity.	Sections 4, 7
FR-06/07	Requirements §5	Persist before forwarding; recover from temporary backend failure.	Sections 4, 7, 10
FR-08/09	Requirements §5	Idempotency and per-item batch validation.	Sections 4, 7
FR-10/11/12	Requirements §5	Current status, history and stale connection behavior.	Sections 5, 7, 8
FR-13/14/15	Requirements §5 and §7	High-temperature rule, alert episode and acknowledgement.	Sections 5, 7, 11
Telemetry processing	Requirements §6	Composite uniqueness, commit boundary, queue/quarantine/retry rules and replay protection.	Sections 4, 7, 8
NFR targets	Requirements §8	Release-test targets for ingest, freshness, query, storage, recovery, restart, retention, observability and usability.	Sections 6, 8, 13
Security baseline	Requirements §9	TLS, credentials, secret hygiene, authz testing, safe errors/logs and dependency review.	Sections 5, 6, 10
AT-01..AT-10	Requirements §10	Mandatory end-to-end acceptance scenarios.	Section 7
M1..M6	Requirements §11	Contract-first week 1 through release evidence week 6.	Sections 4, 13
Definition of Done	Requirements §12	Tests, docs, evidence, no blockers, reproducible setup, ADRs and known limitations.	Sections 7, 11, 13
Sprint 1 QA decisions	QA report §§2-8	Identity, timestamps, batch outcomes and open questions.	Sections 4, 7, 9
Sprint 1 technical decisions	Technical report §§3-11	Immutable duplicate comparison, commit boundary, queue lifecycle, site boundary and exit checklist.	Sections 4, 7, 9

Document Approval

Recommended status: approve as the PO/PM scope-and-demo working baseline after the team confirms the open decisions in Section 9 and verifies that no existing requirement has been unintentionally removed. Changes after approval should follow the change-control rule in Section 13.

Role	Name	Approval / Date
Product Owner / PM		
Technical Lead		
QA Lead		

References

1. Industrial Equipment Monitoring Platform - Requirements Specification, Version 1.0 Draft baseline, Portfolio MVP.
2. QA Sprint 1 - Telemetry Contract, Decision Log and Open Questions, Week 1 / M1 working document.
3. Sprint 01 Technical Report - M1 Contracts and Vertical Slice, Technical Lead Report, 15 September 2026.